

ZARAGOZA AP, Spain

WMO: 081600

Lat: 41.6606N

Lon: 1.0042W

Elev: 846

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.4	31.4	17.3	13.6	39.7	21.4	16.6	39.2	31.9	50.8	29.2	48.2	5.5	290	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	97.9	69.4	94.8	68.7	91.5	67.9	72.0	90.0	70.7	88.1	69.5	86.4	9.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	102.2	76.4	65.4	96.7	76.2	64.0	92.0	76.1	36.2	90.1	35.0	88.3	34.0	86.7	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.4	26.4	23.9	DB	25.1	103.5	3.2	2.7	22.8	105.4	20.9	107.0	19.1	108.6	16.8	110.6	(4)
(5)			WB	23.3	75.3	3.3	1.5	21.0	76.4	19.0	77.3	17.2	78.1	14.8	79.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	44.9	47.1	52.9	57.6	64.9	73.4	77.8	77.4	70.1	62.3	51.3	44.7		
	DBStd	13.18	5.73	5.57	5.55	5.70	6.27	6.52	5.37	5.12	5.39	5.70	5.76	6.36		
	HDD50	560	174	109	34	8	0	0	0	0	0	2	51	182		
	HDD65	2988	622	501	374	229	80	9	1	0	14	120	410	628		
	CDD50	4379	18	29	126	235	462	702	861	848	604	384	91	19		
	CDD65	1329	0	0	1	7	77	260	396	383	168	37	0	0		
	CDH74	13154	0	0	12	118	775	2711	4201	3807	1311	218	1	0		
	CDH80	6059	0	0	0	17	241	1297	2166	1868	437	33	0	0		
(14) Wind	WSAvg	10.3	10.3	11.4	11.0	10.9	10.7	10.7	11.2	10.4	9.5	8.6	10.0	8.9		
(15) Precipitation	PrecAvg	12.4	0.9	0.9	0.9	1.4	1.4	1.2	0.6	0.7	1.0	1.4	1.3	0.8		
	PrecMax	21.3	3.2	2.8	2.5	5.0	5.6	3.9	2.3	3.5	4.0	4.6	4.1	3.0		
	PrecMin	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.0	0.8	0.6	0.7	1.1	1.1	1.0	0.6	0.7	1.0	1.1	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	66.6	75.5	82.3	91.4	100.7	101.0	100.8	93.4	84.1	70.1	64.0		
		MCWB	53.2	53.2	56.3	60.3	65.6	69.1	70.2	70.2	68.3	64.1	58.0	54.0		
	2%	DB	59.4	63.0	71.5	77.2	86.1	95.3	97.8	97.1	88.3	79.4	66.0	59.4		
		MCWB	51.0	51.2	54.7	58.4	63.4	67.9	69.8	69.2	66.6	62.6	56.2	51.5		
	5%	DB	57.0	60.4	67.8	73.5	82.2	91.6	94.9	93.5	84.8	75.7	62.7	56.9		
		MCWB	49.3	49.9	53.5	56.9	61.9	67.0	69.2	68.7	65.6	61.4	53.8	49.8		
	10%	DB	53.9	57.2	64.3	69.8	78.4	87.9	91.4	89.9	81.6	72.2	60.4	53.8		
		MCWB	47.2	48.5	51.9	55.3	60.5	65.7	68.2	68.0	64.1	59.8	52.7	48.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	55.2	58.5	61.9	67.3	72.5	73.9	73.9	71.1	67.1	60.1	55.0		
		MCDB	61.7	63.6	71.4	77.3	86.5	93.4	94.4	92.4	85.4	77.1	67.2	61.1		
	2%	WB	52.0	52.5	56.3	59.8	64.8	70.0	72.0	72.1	69.3	65.1	57.5	52.4		
		MCDB	58.4	60.8	67.8	74.2	81.6	89.5	91.2	88.8	82.8	74.6	63.1	57.8		
	5%	WB	50.2	50.5	54.5	58.0	63.1	68.4	70.7	70.8	67.6	63.5	55.6	50.7		
		MCDB	55.0	58.1	65.2	71.0	78.1	87.2	89.2	86.9	79.8	71.8	60.5	55.3		
	10%	WB	48.4	49.1	52.7	56.2	61.6	66.9	69.4	69.5	66.0	61.8	54.0	49.1		
		MCDB	53.0	56.0	62.3	67.3	75.0	84.0	87.3	85.3	77.9	69.5	58.6	53.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	16.8	19.9	20.2	21.7	24.4	25.1	23.7	20.7	18.1	14.4	13.3		
		MCDBR	17.2	21.5	25.8	26.5	27.9	29.7	29.4	28.3	25.0	22.4	17.2	16.2		
		MCWBR	10.7	12.0	13.3	12.4	11.5	11.1	10.8	10.2	10.0	10.1	9.9	10.3		
		MCDBR	14.9	18.8	22.8	23.9	25.0	27.6	27.4	25.5	22.4	19.2	14.8	14.5		
	5% WB	MCWBR	10.3	11.4	12.3	12.0	11.1	11.2	10.7	10.6	9.6	9.3	9.5	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.313	0.354	0.375	0.390	0.400	0.388	0.393	0.373	0.350	0.316	0.298		
	taud		2.492	2.451	2.354	2.311	2.302	2.302	2.326	2.315	2.371	2.443	2.490	2.499		
	Ebn at Noon		272	285	283	284	281	277	279	275	273	267	262	262		
	Edh at Noon		23	28	35	39	41	41	40	39	34	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		624	927	1323	1674	2011	2263	2332	2021	1560	1061	696	548		
	RadStd		60	100	130	141	122	83	69	74	63	79	68	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.67	N/A	N/A	N/A	N/A	+184	+166
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page